Streamlining Ticket Assignment for Efficient Support Operations

Date: 02/11/2025
Team ID: NM2025TMID05937
Maximum Marks: 4 Marks

PROJECT DESIGN PHASE 2

TECHNOLOGY STACK (ARCHITECTURE & STACK)

The technical architecture for the Streamlining Ticket Assignment for Efficient Support Operations project is designed as a modular, scalable, and secure system to ensure efficient automation and data-driven decision-making.
It follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer, supported by intelligent automation and analytics modules.

Components and Technologies

1. 1. Presentation Layer

- User Interface (UI)
- Dashboards
- Notification Center
- Reporting & Analytics Views

Technologies: HTML5, CSS3, JavaScript, React.js / Angular, REST APIs

2. 2. Application Layer

- Ticket Routing Engine
- Workflow Manager
- Notification Service
- Integration Module

Technologies: Node.js, Python (Flask / Django), Express.js, Java Spring Boot

3. 3. Data Layer

- Ticket Database
- User Database
- Analytics Database
- Logging & Audit Database

Technologies: MySQL, PostgreSQL, MongoDB, Firebase

4. 4. AI & Automation Layer

- NLP Engine
- Machine Learning Model
- Load Balancer

Technologies: Python, TensorFlow, Scikit-learn, spaCy, NLTK

5. 5. Security & Infrastructure Layer

- Authentication & Authorization
- Data Encryption
- Cloud Deployment
- Backup & Recovery

Technologies: OAuth 2.0 / JWT, SSL/TLS, AWS / Azure, Docker, Kubernetes

6. 6. Monitoring & Reporting Layer

- System Monitoring Tools
- Performance Metrics Dashboard
- Log Management

Technologies: Grafana, Kibana, Prometheus, Power BI